



AAPL STOCK PRICE PREDICTION

GREEN MACHINE LEARNING WEB APPLICATION

Ridge Regression Model to Predict Next-Day Adjusted Close Price



Accurate Prediction • Sustainable AI • Real-time Web App • Green & Responsible Technology



1. PROJECT OVERVIEW

Problem Statement

Predict the next-day adjusted closing price of Apple Inc. (AAPL) stock using historical stock data.

Machine Learning Type

Supervised machine learning – Regression, because the target is a continuous price value.

Objective

To develop an accurate and interpretable machine learning model using Ridge Regression and deploy it as an interactive web application with a green and sustainable theme.



2. DATASET



Source Yahoo Finance (yfinance)
AAPL historical data



Period 2010 ~ 2024



Records 3,521 trading days



Target Variable Next Day Adjusted Close Price



Features Used (7) Open, High, Low, Close,
Adj Close, Volume,
Day of Week



3. METHODOLOGY



Data Collection



Data Preprocessing



Model Building



Model Evaluation



Deployment



4. MODEL & RESULTS

Model Used

Ridge Regression
(α (alpha) = 1.0)

$$\hat{y} = X\hat{\beta}$$

where $\hat{y}$ = predicted value, X = feature matrix,
 $\hat{\beta}$ = coefficients with L2 regularization

Model Performance (Test Set)



MAE
1.9049
Lower is better



RMSE
2.7800
Lower is better

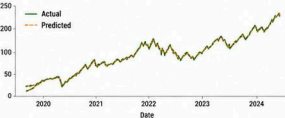


R² Score
0.9990
Closer to 1 is better

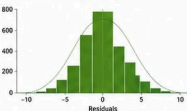


MAPE
4.30%
Lower is better

Actual vs Predicted (Test Set)



Residuals Distribution (Test Set)



Residuals are centered around 0, approximately normally distributed, indicating a good model fit.

5. KEY FINDINGS

- ✓ Ridge Regression with minimal regularization ($\alpha=1.0$) achieves excellent performance.
- ✓ The model achieves a very high R² score of 0.9990, indicating outstanding fit.
- ✓ Predicted prices closely follow actual prices across the test period.
- ✓ The residuals are randomly distributed around zero, validating model reliability.
- ✓ The web application provides an easy-to-use interface for real-time predictions.



6. WEB APPLICATION

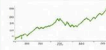
An interactive web app built with Streamlit that allows users to:

- ✓ View historical AAPL stock data
- ✓ Visualize trends and performance
- ✓ Input the latest data to get next-day price prediction
- ✓ Green-themed UI promoting sustainability and clarity

AAPL Stock Price Prediction

Prediction Result

Predicted Next-Day Adjusted Close Price
\$231.47
(Example)



7. TRY THE APP LIVE!

Scan the QR code below to access the live demo!



SCAN ME

Or visit the link:

<https://aapl-stock-prediction-3scusseltfmmzjcthk2lp8.streamlit.app/>

GREEN AI FOR A BETTER FUTURE

Leveraging machine learning for smarter investments and a more sustainable tomorrow.



TECHNOLOGIES USED



Python

Streamlit

Scikit-learn

yfinance

DEVELOPED BY

CSM3601 – Machine Learning Project
Universiti Putra Malaysia

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"Predicting today for a smarter, greener tomorrow." 🌱